

# Influence of storage and household processing on the agaritine content of the cultivated *Agaricus* mushroom.

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Agaritine (N-(gamma-L(+)-glutamyl)-4-hydroxymethyl-phenylhydrazine) was identified and quantified by high-pressure liquid chromatography and used as a marker for the occurrence of phenylhydrazine derivatives in the cultivated *Agaricus bitorquis* and *A. garicus hortensis* mushrooms. Although relatively high levels of agaritine (around 700 mg kg<sup>-1</sup>) could be found in freshly harvested *A. bitorquis* from early flushes, samples from supermarkets contained less agaritine. The content of 28 samples varied between 165 and 457 mg kg<sup>-1</sup>, on average being 272 +/- 69 mg kg<sup>-1</sup>. The highest amounts of agaritine were found in the skin of the cap and in the gills, the lowest being in the stem. There was no significant difference in agaritine content of the two mushroom species in our study. Pronounced reduction in agaritine content was observed during storage of mushrooms in the refrigerator or freezer, as well as during drying of the mushrooms. The degree of reduction was dependent on the length and condition of storage and was usually in the region 20-75%. No reduction in agaritine content was observed during freeze-drying. Depending on the cooking procedure, household processing of cultivated *Agaricus* mushrooms reduced the agaritine content to various degrees. Boiling extracted around 50% of the agaritine content into the cooking broth within 5min and degraded 20-25% of the original agaritine content of the mushrooms. Prolonged boiling, as when preparing a sauce, reduced the content in the solid mushroom further (around 10% left after 2h). Dry baking of the cultivated mushroom, a process similar to pizza baking, reduced the agaritine content by approximately 25%, whereas frying in oil or butter or deep frying resulted in a more marked reduction (35-70%). Microwave processing of the cultivated mushrooms reduced the agaritine content to one-third of the original level. Thus, the exposure to agaritine was substantially less when consuming processed *Agaricus* mushrooms as compared with consuming the raw mushrooms. However, it is not yet known to what extent agaritine and other phenylhydrazine derivatives occurring in the cultivated mushroom are degraded into other biologically active compounds during the cooking procedure.